Purpose

TEL-0 — telemetry/ Purpose + boundaries

Piece A — Purpose

File: `Axion/libraries/telemetry/TEL-0_purpose.md`

---
library: telemetry
id: TEL-0
schema_version: 1.0.0
status: draft

TEL-0 — telemetry/ Purpose + Boundaries

Purpose
`telemetry/` defines Axion's **event and metrics contracts**:
- what events exist (event types + required fields)
- what run metrics exist (durations, counts, cost signals)
- where telemetry is allowed to be sent (sink policies)
- privacy/safety constraints for telemetry payloads

Telemetry helps measure pipeline health without changing pipeline semantics.

What it governs (in scope)
- Telemetry event schemas (per event type)
- Run metrics schema (aggregate metrics per run/stage)
- Telemetry sink policy (allowed destinations, redaction rules)

What it does NOT govern (out of scope)
- Operational monitoring/alert thresholds → `ops/`
- Logging/tracing standards → `ops/`
- Gate DSL → `gates/`
- Orchestration mechanics → `orchestration/`

Consumers
- Pipeline runtime (emits events)
- Operator UI (dashboards, run summaries)
- Cost modeling (ops) (reads telemetry)

Determinism requirements
- Telemetry must not affect run outcomes (no feedback loops in core path).

- Event schemas are versioned and pinned for compatibility.

Piece B — Boundary checklist

File: `Axion/libraries/telemetry/TEL-0_boundary_checklist.md`

```
---
library: telemetry
id: TEL-0a
schema_version: 1.0.0
status: draft
---

# TEL-0a — Boundary Checklist

Belongs in `telemetry/` if it answers:
- “What events do we emit and what fields do they have?”
- “What metrics do we compute?”
- “Where can telemetry go, and what must be redacted?”

Does NOT belong in `telemetry/` if it answers:
- “What thresholds trigger alerts?” (ops)
- “How do gates pass/fail?” (gates)
```

Telemetry event schemas

TEL-1 — Event schemas (event type registry + base event contract)

Piece A — Base telemetry event model (concept)

File: `Axion/libraries/telemetry/TEL-1_event_model.md`

```
---
library: telemetry
id: TEL-1
schema_version: 1.0.0
status: draft
---

# TEL-1 — Telemetry Event Model

## Purpose
Define a base event structure that all telemetry events share so event ingestion is uniform.

## Base event fields (minimum)
- event_id
- event_type
- schema_version
- run_id
- stage_id (optional for run-wide events)
- timestamp
- payload (event-specific fields)

## Design rule
Event payload must be safe-by-default:
- no secrets
- no raw user PII
- no full document content
Use references/ids + redaction rules.
```

Piece B — Base event schema (JSON Schema)

File:
`Axion/libraries/telemetry/schemas/telemetry_event_base.v1.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "schema_version": "1.0.0",
  "id": "telemetry_event_base.v1",
  "type": "object",
  "required": ["event_id", "event_type", "schema_version", "run_id", "timestamp", "payload"],
  "properties": {
    "event_id": { "type": "string", "pattern": "^EVT-[A-Z0-9]{6,}$" },
    "event_type": { "type": "string", "minLength": 3 },
    "schema_version": { "type": "string", "minLength": 1 },
    "run_id": { "type": "string" },
    "stage_id": { "type": "string" },
    "timestamp": { "type": "string", "format": "date-time" },
    "payload": { "type": "object", "additionalProperties": true }
  },
  "additionalProperties": false
}
```

Piece C — Event types registry schema

File:
`Axion/libraries/telemetry/schemas/telemetry_event_types.v1.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "schema_version": "1.0.0",
  "id": "telemetry_event_types.v1",
  "type": "object",
  "required": ["registry_id", "schema_version", "types"],
  "properties": {
    "registry_id": { "type": "string", "minLength": 3 },
    "schema_version": { "type": "string", "minLength": 1 },
    "types": {
      "type": "array",
      "minItems": 1,
      "items": {
        "type": "object",
        "required": ["event_type", "schema_ref", "description"],
        "properties": {
          "event_type": { "type": "string", "minLength": 3 },
          "schema_ref": { "type": "string", "minLength": 3 },
          "description": { "type": "string", "minLength": 5 }
```

```
      },
      "additionalProperties": false
    }
  }
},
"additionalProperties": false
}
```

Piece D — Starter event types registry

File: `Axion/libraries/telemetry/registries/telemetry_event_types.v1.json`

```
{
  "schema_version": "1.0.0",
  "registry_id": "telemetry_event_types.v1",
  "types": [
    {
      "event_type": "run.started",
      "schema_ref": "telemetry_event_base.v1",
      "description": "Run started event."
    },
    {
      "event_type": "stage.started",
      "schema_ref": "telemetry_event_base.v1",
      "description": "Stage started event."
    },
    {
      "event_type": "stage.ended",
      "schema_ref": "telemetry_event_base.v1",
      "description": "Stage ended event (duration and status)."
    },
    {
      "event_type": "gate.evaluated",
      "schema_ref": "telemetry_event_base.v1",
      "description": "Gate evaluation result."
    },
    {
      "event_type": "run.ended",
      "schema_ref": "telemetry_event_base.v1",
      "description": "Run ended event."
    }
  ]
```

}

Piece E — Determinism rules

File: `Axion/libraries/telemetry/TEL-1_determinism_rules.md`

```
---
library: telemetry
id: TEL-1a
schema_version: 1.0.0
status: draft
---

# TEL-1a — Determinism Rules (Events)

- event_type must exist in telemetry_event_types registry (strict mode).
- base event schema is pinned per run version.
- telemetry emission must not affect run decisions (fire-and-forget).
- payload must comply with sink redaction rules (TEL-3).
```

Piece F — Validation checklist

File: `Axion/libraries/telemetry/TEL-1_validation_checklist.md`

```
---
library: telemetry
id: TEL-1b
schema_version: 1.0.0
status: draft
---

# TEL-1b — Validation Checklist

- [ ] telemetry_event_base validates
- [ ] telemetry_event_types registry validates and is pinned
- [ ] each event_type references a known schema_ref
- [ ] event payloads avoid secrets/PII by policy
```

Run metrics schemas

TEL-2 — Run metrics schema (aggregate metrics per run/stage)

Piece A — Metrics model (concept)

File: `Axion/libraries/telemetry/TEL-2_run_metrics_model.md`

```
---
library: telemetry
id: TEL-2
schema_version: 1.0.0
status: draft
---

# TEL-2 — Run Metrics Model

## Purpose
Define deterministic aggregate metrics for:
- whole runs
- stages
- gates

These metrics support operator visibility and cost/perf analysis.

## Metric categories (minimum)
- timing: durations per stage and total
- counts: artifacts produced, templates selected, proofs recorded
- outcomes: pass/fail per gate, final run status
- resource signals (optional): tokens, cost, compute time

## Design rules
- Metrics are derived from pinned run artifacts/events.
- Metrics output is small and structured.
```

Piece B — Run metrics schema (JSON Schema)

File: `Axion/libraries/telemetry/schemas/run_metrics.v1.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
```

```
"schema_version": "1.0.0",
"id": "run_metrics.v1",
"type": "object",
"required": ["metrics_id", "run_id", "computed_at", "run", "stages", "gates"],
"properties": {
  "metrics_id": { "type": "string", "pattern": "^MET-[A-Z0-9]{6,}$" },
  "run_id": { "type": "string" },
  "computed_at": { "type": "string", "format": "date-time" },

  "run": {
    "type": "object",
    "required": ["status"],
    "properties": {
      "status": { "type": "string", "enum": ["succeeded", "failed", "paused"] },
      "duration_ms": { "type": "integer", "minimum": 0 },
      "tokens_in": { "type": "integer", "minimum": 0 },
      "tokens_out": { "type": "integer", "minimum": 0 },
      "cost_usd": { "type": "number", "minimum": 0 }
    },
    "additionalProperties": false
  },

  "stages": {
    "type": "array",
    "items": {
      "type": "object",
      "required": ["stage_id", "status"],
      "properties": {
        "stage_id": { "type": "string" },
        "status": { "type": "string", "enum": ["succeeded", "failed", "skipped"] },
        "duration_ms": { "type": "integer", "minimum": 0 },
        "artifacts_produced": { "type": "integer", "minimum": 0 }
      },
      "additionalProperties": false
    }
  },

  "gates": {
    "type": "array",
    "items": {
      "type": "object",
      "required": ["gate_id", "status"],
      "properties": {
        "gate_id": { "type": "string" },
```

```
          "status": { "type": "string", "enum": ["pass", "fail", "warn"] }
        },
        "additionalProperties": false
      }
    }
  },
  "additionalProperties": false
}
```

Piece C — Determinism rules (metrics computation)

File: `Axion/libraries/telemetry/TEL-2_determinism_rules.md`

```
---
library: telemetry
id: TEL-2a
schema_version: 1.0.0
status: draft
---

# TEL-2a — Determinism Rules (Metrics)

- Metrics are derived from pinned run artifacts:
  - run manifest timestamps
  - stage reports
  - gate reports
  - proof ledger counts
  - template selection counts
- Ordering:
  - stages sorted by stage order (or stage_id lexicographic if unavailable)
  - gates sorted by gate order (or gate_id lexicographic)
- Optional token/cost metrics must specify their source (adapter/runtime) if present.
```

Piece D — Validation checklist

File: `Axion/libraries/telemetry/TEL-2_validation_checklist.md`

```
---
library: telemetry
id: TEL-2b
```

schema_version: 1.0.0
status: draft

TEL-2b — Validation Checklist

- [ ] run_metrics validates against schema
- [ ] durations are non-negative and consistent
- [ ] stage/gate ordering is deterministic
- [ ] token/cost fields omitted if unavailable (no fake data)

Sink policies

TEL-3 — Telemetry sink policy (destinations + redaction)

Piece A — Sink policy model (concept)

File: `Axion/libraries/telemetry/TEL-3_sink_policy_model.md`

```
---
library: telemetry
id: TEL-3
schema_version: 1.0.0
status: draft
---

# TEL-3 — Telemetry Sink Policy Model

## Purpose
Define:
- where telemetry is allowed to be sent (sinks)
- what fields must be redacted
- what payload classes are forbidden

This is the safety layer for telemetry.

## Core rules
- Telemetry must never include secrets.
- Telemetry must never include full doc content (templates, canonical spec prose).
- Prefer counts + ids + hashes over raw content.
- External sinks require stricter redaction.

## Sink types (examples)
- local_file (dev)
- internal_http (internal collector)
- external_http (third-party analytics) — restricted
```

Piece B — Sink policy schema (JSON Schema)

File: `Axion/libraries/telemetry/schemas/telemetry_sink_policy.v1.schema.json`

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
```

```
  "schema_version": "1.0.0",
  "id": "telemetry_sink_policy.v1",
  "type": "object",
  "required": ["policy_id", "schema_version", "sinks", "redaction"],
  "properties": {
    "policy_id": { "type": "string", "pattern": "^TELPOL-[A-Z0-9]{6,}$" },
    "schema_version": { "type": "string", "minLength": 1 },

    "sinks": {
      "type": "array",
      "minItems": 1,
      "items": {
        "type": "object",
        "required": ["sink_id", "type", "enabled"],
        "properties": {
          "sink_id": { "type": "string", "minLength": 3 },
          "type": { "type": "string", "enum": ["local_file", "internal_http", "external_http"] },
          "enabled": { "type": "boolean" },
          "endpoint": { "type": "string" },
          "notes": { "type": "string" }
        },
        "additionalProperties": false
      }
    },

    "redaction": {
      "type": "object",
      "required": ["deny_keys", "deny_patterns", "external_strict_mode"],
      "properties": {
        "deny_keys": { "type": "array", "items": { "type": "string" } },
        "deny_patterns": { "type": "array", "items": { "type": "string" } },
        "external_strict_mode": { "type": "boolean" }
      },
      "additionalProperties": false
    }
  },
  "additionalProperties": false
}
```

Piece C — Starter sink policy registry

File: `Axion/libraries/telemetry/registries/telemetry_sink_policy.v1.json`

```
{
  "schema_version": "1.0.0",
  "policy_id": "TELPOL-BASE01",
  "sinks": [
    {
      "sink_id": "sink-local-dev",
      "type": "local_file",
      "enabled": true,
      "endpoint": "telemetry/events.jsonl",
      "notes": "Local dev sink only."
    },
    {
      "sink_id": "sink-internal-collector",
      "type": "internal_http",
      "enabled": false,
      "endpoint": "http://internal-collector/ingest",
      "notes": "Internal telemetry collector (if deployed)."
    },
    {
      "sink_id": "sink-external",
      "type": "external_http",
      "enabled": false,
      "endpoint": "",
      "notes": "External sinks disabled by default."
    }
  ],
  "redaction": {
    "deny_keys": [
      "api_key",
      "token",
      "password",
      "secret",
      "authorization",
      "cookie",
      "session",
      "private_key",
      "access_key"
    ],
    "deny_patterns": [
      "(?i)bearer\\s+[a-z0-9\\-\\._]+",
      "(?i)api[_-]?key\\s*[:=]\\s*.+",
      "(?i)secret\\s*[:=]\\s*.+"
    ],
    "external_strict_mode": true
```

```
  }
}
```

Piece D — Determinism rules (redaction + sink routing)

File: `Axion/libraries/telemetry/TEL-3_determinism_rules.md`

```
---
library: telemetry
id: TEL-3a
schema_version: 1.0.0
status: draft
---

# TEL-3a — Determinism Rules (Telemetry Sinks)

- Sink selection is derived from pinned sink policy registry.
- Redaction is applied deterministically:
  - remove deny_keys anywhere in payload (deep traversal)
  - remove/replace values matching deny_patterns
- External sinks require external_strict_mode == true:
  - drop any event with unknown payload keys (optional strict behavior)
  - enforce minimal payload whitelist per event_type (if implemented later)
```

Piece E — Validation checklist

File: `Axion/libraries/telemetry/TEL-3_validation_checklist.md`

```
---
library: telemetry
id: TEL-3b
schema_version: 1.0.0
status: draft
---

# TEL-3b — Validation Checklist

- [ ] telemetry_sink_policy validates against schema
- [ ] deny_keys and deny_patterns exist and cover secrets
- [ ] external sinks disabled by default
```

- [ ] redaction is deterministic and applied before routing
- [ ] telemetry never includes full doc content

PII/redaction rules

TEL-4 — PII / redaction rules (privacy)

Piece A — Privacy model (concept)

File: `Axion/libraries/telemetry/TEL-4_privacy_model.md`

---
library: telemetry
id: TEL-4
schema_version: 1.0.0
status: draft

TEL-4 — Privacy Model (PII + Redaction)

Purpose
Define what counts as sensitive data in telemetry and how it must be handled:
- PII (personally identifiable information)
- secrets/credentials
- proprietary content (full docs, code dumps, private business data)

Privacy stance
Telemetry is **metadata-only by default**:
- ids, counts, timings, statuses
- hashes (optional)
- references/paths (internal only)

Never emit:
- full document content
- raw chat transcripts
- raw intake free-text (unless explicitly allowlisted and redacted)

Data classes
Class A — Secrets (always forbidden)
Examples:
- API keys, tokens, passwords, session cookies
- Authorization headers
- Private keys, signing secrets

Class B — PII (forbidden unless explicitly allowlisted)
Examples:
- names, emails, phone numbers

- physical addresses
- government IDs
- biometric identifiers

Class C — Sensitive business content (forbidden by default)
Examples:
- customer lists
- pricing sheets
- internal strategy docs
- non-public architecture diagrams in full text

Class D — Safe operational metadata (allowed)
Examples:
- run_id, stage_id, gate_id
- durations
- counts of artifacts/proofs/templates
- success/fail status

Piece B — Redaction rules (operational)

File: `Axion/libraries/telemetry/TEL-4_redaction_rules.md`

---
library: telemetry
id: TEL-4a
schema_version: 1.0.0
status: draft

TEL-4a — Redaction Rules

Redaction actions (in order)
1) **Drop** forbidden keys (deep traversal)
2) **Scrub** forbidden patterns in string values
3) **Truncate** high-risk free-text fields (if allowlisted) to max length
4) **Hash** certain identifiers when required for correlation
5) **Block** event emission if payload still violates policy (external sinks)

Key-based deny list (minimum)
Drop any key matching (case-insensitive):
- api_key, token, password, secret, authorization, cookie, session
- private_key, access_key, refresh_token, id_token

Pattern-based deny list (minimum)
Scrub any value matching:
- bearer tokens
- key=value secrets
- PEM blocks
- JWT-like patterns

Free-text handling
If an event payload includes any free-text field (notes, message, summary):
- internal sinks: allow only if allowlisted + truncated
- external sinks: block unless explicitly allowlisted and scrubbed

“No content dumps”
If payload contains fields like:
- full_document
- raw_markdown
- full_code
Then:
- internal sinks: drop field
- external sinks: drop field and mark event as reduced

Piece C — Machine-readable privacy policy registry

File:
`Axion/libraries/telemetry/registries/telemetry_privacy_policy.v1.json`

```
{
 "schema_version": "1.0.0",
 "policy_id": "TELPRIV-BASE01",
 "data_classes": {
   "forbidden_always": ["secrets"],
   "forbidden_by_default": ["pii", "sensitive_business", "content_dumps"],
   "allowed": ["operational_metadata"]
 },
 "deny_keys": [
   "api_key",
   "token",
   "password",
   "secret",
   "authorization",
   "cookie",
   "session",
   "private_key",
```

```
  "access_key",
  "refresh_token",
  "id_token"
 ],
 "deny_patterns": [
  "(?i)bearer\\s+[a-z0-9\\-\\._]+",
  "(?i)api[_-]?key\\s*[:=]\\s*.+",
  "(?i)secret\\s*[:=]\\s*.+",
  "-----BEGIN[\\s\\S]+?-----END[\\s\\S]+?-----"
 ],
 "free_text": {
  "internal_allowlist_keys": ["reason", "error_summary"],
  "max_chars_internal": 280,
  "external_allowed": false
 },
 "external_sink_rules": {
  "block_on_unknown_keys": true,
  "block_on_free_text": true
 }
}
```

Piece D — Privacy policy schema

File: `Axion/libraries/telemetry/schemas/telemetry_privacy_policy.v1.schema.json`

```
{
 "$schema": "https://json-schema.org/draft/2020-12/schema",
 "schema_version": "1.0.0",
 "id": "telemetry_privacy_policy.v1",
 "type": "object",
 "required": ["policy_id", "schema_version", "deny_keys", "deny_patterns", "free_text",
"external_sink_rules"],
 "properties": {
  "policy_id": { "type": "string", "minLength": 6 },
  "schema_version": { "type": "string", "minLength": 1 },

  "deny_keys": { "type": "array", "items": { "type": "string" } },
  "deny_patterns": { "type": "array", "items": { "type": "string" } },

  "free_text": {
   "type": "object",
```

```
    "required": ["internal_allowlist_keys", "max_chars_internal", "external_allowed"],
    "properties": {
      "internal_allowlist_keys": { "type": "array", "items": { "type": "string" } },
      "max_chars_internal": { "type": "integer", "minimum": 0 },
      "external_allowed": { "type": "boolean" }
    },
    "additionalProperties": false
  },

  "external_sink_rules": {
    "type": "object",
    "required": ["block_on_unknown_keys", "block_on_free_text"],
    "properties": {
      "block_on_unknown_keys": { "type": "boolean" },
      "block_on_free_text": { "type": "boolean" }
    },
    "additionalProperties": false
  }
},
"additionalProperties": false
}
```

Piece E — Determinism rules (privacy)

File: `Axion/libraries/telemetry/TEL-4_determinism_rules.md`

```
---
library: telemetry
id: TEL-4b
schema_version: 1.0.0
status: draft
---

# TEL-4b — Determinism Rules (Privacy)

- Privacy policy registry is pinned.
- Redaction is deterministic (same input → same output).
- External sink policy is stricter than internal.
- If a payload violates policy after redaction:
 - internal sinks: drop violating fields and emit reduced event
 - external sinks: block event emission
```

Piece F — Validation checklist

File: `Axion/libraries/telemetry/TEL-4_validation_checklist.md`

---
library: telemetry
id: TEL-4c
schema_version: 1.0.0
status: draft

TEL-4c — Validation Checklist (Privacy)

- [ ] telemetry_privacy_policy schema exists and validates
- [ ] deny_keys + deny_patterns cover secrets
- [ ] free-text rules are explicit (allowlist + max chars)
- [ ] external sinks block free-text by default
- [ ] no content dumps are allowed

Gates

TEL-4 — Telemetry gates (policy compliance + schema validity)

Piece A — Telemetry gates intent

File: `Axion/libraries/telemetry/TEL-4_telemetry_gates.md`

---
library: telemetry
id: TEL-4
schema_version: 1.0.0
status: draft

TEL-4 — Telemetry Gates

Purpose
Ensure telemetry is schema-valid, safe, and policy compliant.

Gate set (minimum)

TEL-GATE-01 — Event types registry pinned + valid
- telemetry_event_types registry pinned
- validates against telemetry_event_types.v1

TEL-GATE-02 — Emitted events validate against base schema
- telemetry events validate against telemetry_event_base.v1
- event_type values exist in telemetry_event_types (strict mode)

TEL-GATE-03 — Sink policy pinned + valid
- telemetry_sink_policy registry pinned
- validates against telemetry_sink_policy.v1

TEL-GATE-04 — Redaction compliance
- deny_keys removed
- deny_patterns scrubbed
- external_strict_mode enforced for external sinks

TEL-GATE-05 — Telemetry non-interference
- telemetry failures must not fail the run by default
- if strict mode enabled, failures can warn/pause (policy controlled)

Piece B — Machine-readable gate spec (optional)

File: `Axion/libraries/telemetry/TEL-4_telemetry_gates.spec.json`


```
{
  "schema_version": "1.0.0",
  "contract_id": "TEL-4a",
  "gates": [
    { "gate_id": "TEL-GATE-01", "name": "Event types registry pinned + valid", "severity": "hard_stop", "checks": ["pinned(telemetry_event_types)", "schema_valid(telemetry_event_types.v1)"] },
    { "gate_id": "TEL-GATE-02", "name": "Emitted events validate", "severity": "warn", "checks": ["events schema_valid(telemetry_event_base.v1)", "event_type exists in registry"] },
    { "gate_id": "TEL-GATE-03", "name": "Sink policy pinned + valid", "severity": "hard_stop", "checks": ["pinned(telemetry_sink_policy)", "schema_valid(telemetry_sink_policy.v1)"] },
    { "gate_id": "TEL-GATE-04", "name": "Redaction compliance", "severity": "hard_stop", "checks": ["deny_keys removed", "deny_patterns scrubbed", "external_strict_mode enforced"] },
    { "gate_id": "TEL-GATE-05", "name": "Non-interference", "severity": "policy_controlled", "checks": ["telemetry failures do not fail run unless strict policy"] }
  ]
}
```

Piece C — Determinism rules

File: `Axion/libraries/telemetry/TEL-4_determinism_rules.md`


```
---
library: telemetry
id: TEL-4b
schema_version: 1.0.0
status: draft
---
```


TEL-4b — Determinism Rules (Telemetry Gates)

- Registry and sink policy are pinned.
- Redaction algorithm is deterministic (same input → same output).
- Gate evaluation order is fixed TEL-GATE-01..05.
- Telemetry failures default to warn; strict mode is policy controlled.

Piece D — Validation checklist

File: `Axion/libraries/telemetry/TEL-4_validation_checklist.md`

---
library: telemetry
id: TEL-4c
schema_version: 1.0.0
status: draft

TEL-4c — Validation Checklist

- [ ] Telemetry gates cover registry validity, event schema validity, sink policy, redaction, non-interference
- [ ] Telemetry failures cannot silently leak secrets
- [ ] Deterministic ordering and deterministic redaction enforced

Minimum viable telemetry

TEL-6 — Minimum viable telemetry set (runtime-required files + “done”)

Piece A — MV telemetry inventory

File: `Axion/libraries/telemetry/TEL-6_minimum_viable_set.md`

---
library: telemetry
id: TEL-6
schema_version: 1.0.0
status: draft

TEL-6 — Minimum Viable telemetry/ Set

Goal
Define the smallest telemetry library set required to emit safe, schema-valid events and compute run metrics without leaking secrets/PII.

Required runtime files (minimum)
1) Core docs
- TEL-0_purpose.md
- TEL-0_boundary_checklist.md
- TEL-1_event_model.md
- TEL-1_determinism_rules.md
- TEL-1_validation_checklist.md
- TEL-2_run_metrics_model.md
- TEL-2_determinism_rules.md
- TEL-2_validation_checklist.md
- TEL-3_sink_policy_model.md
- TEL-3_determinism_rules.md
- TEL-3_validation_checklist.md
- TEL-4_privacy_model.md
- TEL-4_redaction_rules.md
- TEL-4_determinism_rules.md
- TEL-4_validation_checklist.md
- TEL-5_telemetry_gates.md
- TEL-5_determinism_rules.md
- TEL-5_validation_checklist.md
- TEL-6_minimum_viable_set.md

2) Schemas (must be in schema_registry)
- schemas/telemetry_event_base.v1.schema.json
- schemas/telemetry_event_types.v1.schema.json
- schemas/run_metrics.v1.schema.json
- schemas/telemetry_sink_policy.v1.schema.json
- schemas/telemetry_privacy_policy.v1.schema.json

3) Registries (runtime defaults)
- registries/telemetry_event_types.v1.json
- registries/telemetry_sink_policy.v1.json
- registries/telemetry_privacy_policy.v1.json

4) Gate specs (optional)
- TEL-5_telemetry_gates.spec.json

Piece B — Definition of done (telemetry/)

File: `Axion/libraries/telemetry/TEL-6_definition_of_done.md`

```
---
library: telemetry
id: TEL-6a
schema_version: 1.0.0
status: draft
---

# TEL-6a — telemetry/ Definition of Done

telemetry/ is “done” when:

## Schemas + registries
- [ ] All telemetry schemas validate (JSON Schema check)
- [ ] event types registry validates and is pinned
- [ ] sink policy validates and is pinned
- [ ] privacy policy validates and is pinned

## Runtime behavior (contract-level)
- [ ] Events emitted validate against telemetry_event_base
- [ ] event_type values are registry-backed (strict mode optional)
- [ ] Redaction is applied deterministically before sink routing
- [ ] External sinks are disabled by default and block free-text by default
- [ ] Run metrics can be computed deterministically from pinned artifacts/events
```

Safety
- [ ] No secrets or tokens are emitted
- [ ] No PII is emitted by default
- [ ] No full content dumps are emitted (docs/code/intake text)
- [ ] Violating events are dropped (external) or reduced (internal) per policy

Non-interference
- [ ] Telemetry failures do not fail the run by default (policy-controlled strict mode only)

Piece C — Minimal folder tree

File: `Axion/libraries/telemetry/TEL-6_minimal_tree.txt`

```
Axion/libraries/telemetry/
 TEL-0_purpose.md
 TEL-0_boundary_checklist.md
 TEL-1_event_model.md
 TEL-1_determinism_rules.md
 TEL-1_validation_checklist.md
 TEL-2_run_metrics_model.md
 TEL-2_determinism_rules.md
 TEL-2_validation_checklist.md
 TEL-3_sink_policy_model.md
 TEL-3_determinism_rules.md
 TEL-3_validation_checklist.md
 TEL-4_privacy_model.md
 TEL-4_redaction_rules.md
 TEL-4_determinism_rules.md
 TEL-4_validation_checklist.md
 TEL-5_telemetry_gates.md
 TEL-5_telemetry_gates.spec.json
 TEL-5_determinism_rules.md
 TEL-5_validation_checklist.md
 TEL-6_minimum_viable_set.md
 TEL-6_definition_of_done.md
 TEL-6_minimal_tree.txt

 schemas/
   telemetry_event_base.v1.schema.json
   telemetry_event_types.v1.schema.json
   run_metrics.v1.schema.json
   telemetry_sink_policy.v1.schema.json
   telemetry_privacy_policy.v1.schema.json
```

```
registries/
  telemetry_event_types.v1.json
  telemetry_sink_policy.v1.json
  telemetry_privacy_policy.v1.json
```